THE CATEGORY OF ETALE NEIGHBOURHOODS OF A GEOMETRIC POINT IS COFILTERED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author's old-style \not = and the arrow label ^-\Delta in the displayed square are reproduced verbatim.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. Both referenced results live in the same chapter and are located at their \label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Etale neighbourhood of a geometric point of a scheme. Same chapter, the definition introducing an etale neighbourhood (U, ubar) of a geometric point sbar of S, with U etale over S. Supplies the objects of the category whose cofilteredness is asserted; the two constructions are the counted content.
- Permanence of etale morphisms under base change and composition, and between etale schemes. Same chapter, Proposition labelled proposition-etale-morphisms (source line 2592, Stacks tag 03PC) and Remark labelled remark-etale-between-etale (source line 3230, Stacks tag 03PP); invoked at proof lines 3279-3281, 3294-3296 and 3297-3298. Supply base change, composition and the criterion that a morphism between schemes etale over a base is etale; the fibre product constructions that use them are retained in the counted span.

Source reference. Original proposition 03PC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2592–2592. Same-chapter pointer invoked at proof lines 3281 and 3296 for permanence of etale morphisms under base change and composition; the two fibre product constructions and the nonemptyness check are retained verbatim in the counted span.

Source reference. Original remark 03PP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3230–3230. Same-chapter pointer invoked at proof line 3298 for the criterion that a morphism between schemes etale over S is etale, applied to (h_1, h_2); the base change and composition steps that follow are retained in the counted span.

Lemma 0.1. *Let S be a scheme, and let $\overline{s}$ be a geometric point of S. The category of étale neighborhoods is cofiltered. More precisely:*

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(1) *Let $(U_i, \overline{u}_i)_{i=1,2}$ be two étale neighborhoods of $\overline{s}$ in S. Then there exists a third étale neighborhood $(U, \overline{u})$ and morphisms $(U, \overline{u}) \to (U_i, \overline{u}_i)$, $i = 1, 2$.*
(2) *Let $h_1, h_2 : (U, \overline{u}) \to (U', \overline{u}')$ be two morphisms between étale neighborhoods of $\overline{s}$. Then there exist an étale neighborhood $(U'', \overline{u}'')$ and a morphism $h : (U'', \overline{u}'') \to (U, \overline{u})$ which equalizes h_1 and h_2, i.e., such that $h_1 \circ h = h_2 \circ h$.*

Proof. For part (1), consider the fibre product $U = U_1 \times_S U_2$. It is étale over both U_1 and U_2 because étale morphisms are preserved under base change, see Proposition 03PC. The map $\overline{s} \to U$ defined by $(\overline{u}_1, \overline{u}_2)$ gives it the structure of an étale neighborhood mapping to both U_1 and U_2. For part (2), define U'' as the fibre product

$$\begin{array}{ccc} U'' & \longrightarrow & U \\ \downarrow & & \downarrow{\scriptstyle (h_1, h_2)} \\ U' & \xrightarrow{\Delta} & U' \times_S U'. \end{array}$$

Since $\overline{u}$ and $\overline{u}'$ agree over S with $\overline{s}$, we see that $\overline{u}'' = (\overline{u}, \overline{u}')$ is a geometric point of U''. In particular $U'' \neq \emptyset$. Moreover, since U' is étale over S, so is the fibre product $U' \times_S U'$ (see Proposition 03PC). Hence the vertical arrow (h_1, h_2) is étale by Remark 03PP above. Therefore U'' is étale over U' by base change, and hence also étale over S (because compositions of étale morphisms are étale). Thus $(U'', \overline{u}'')$ is a solution to the problem. □